Figure 1.9 Blockchains stored in local file systems

At the time of this writing (2020), Bitcoin block creation (or mining) time--and, hence, Tx confirmation time--is about 10 minutes. On Ethereum, block confirmation takes about 10 to 19 seconds, whereas transaction confirmation time on credit cards takes less than a second. Recall the speed of your internet connection 10 or 20 years back; blockchain technology is experiencing a similar situation in these early years of its existence. The developer community at the blockchain protocol level is working on improving Tx confirmation times by using various consensus algorithms and by using relaying techniques at the network level.

1.3.3 Disintermediation protocol

Like any transportation infrastructure, a blockchain infrastructure has rules that you need to follow. If drivers don’t follow the laws of the road, chaos and gridlock ensue. A protocol or a set of rules governs the structure and operation of a blockchain. A blockchain protocol defines the following, among other things:

- The structure of a blockchain (transactions, blocks, and chain of blocks)
- Fundamental algorithms and standards for encryption, hashing, and state management
- Methods for implementing consensus and a consistent chain of blocks
- Techniques for handling exceptions resulting in an inconsistent ledger